California College of the Arts

Systems Design

March 10, 2026

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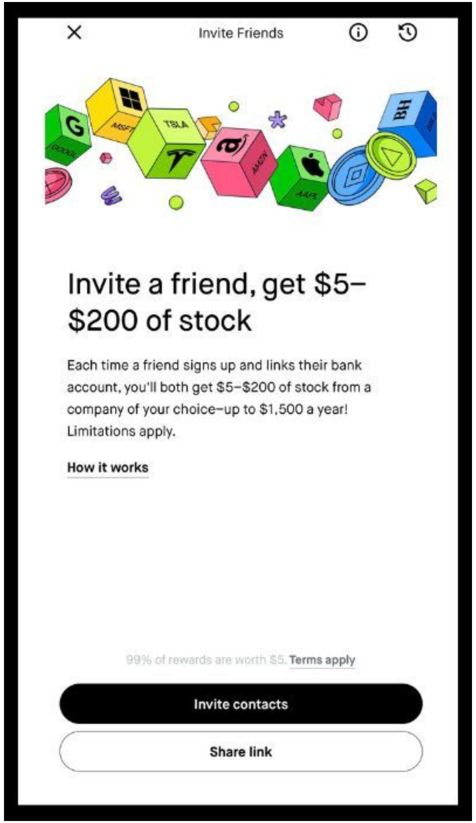
Systems Redesign

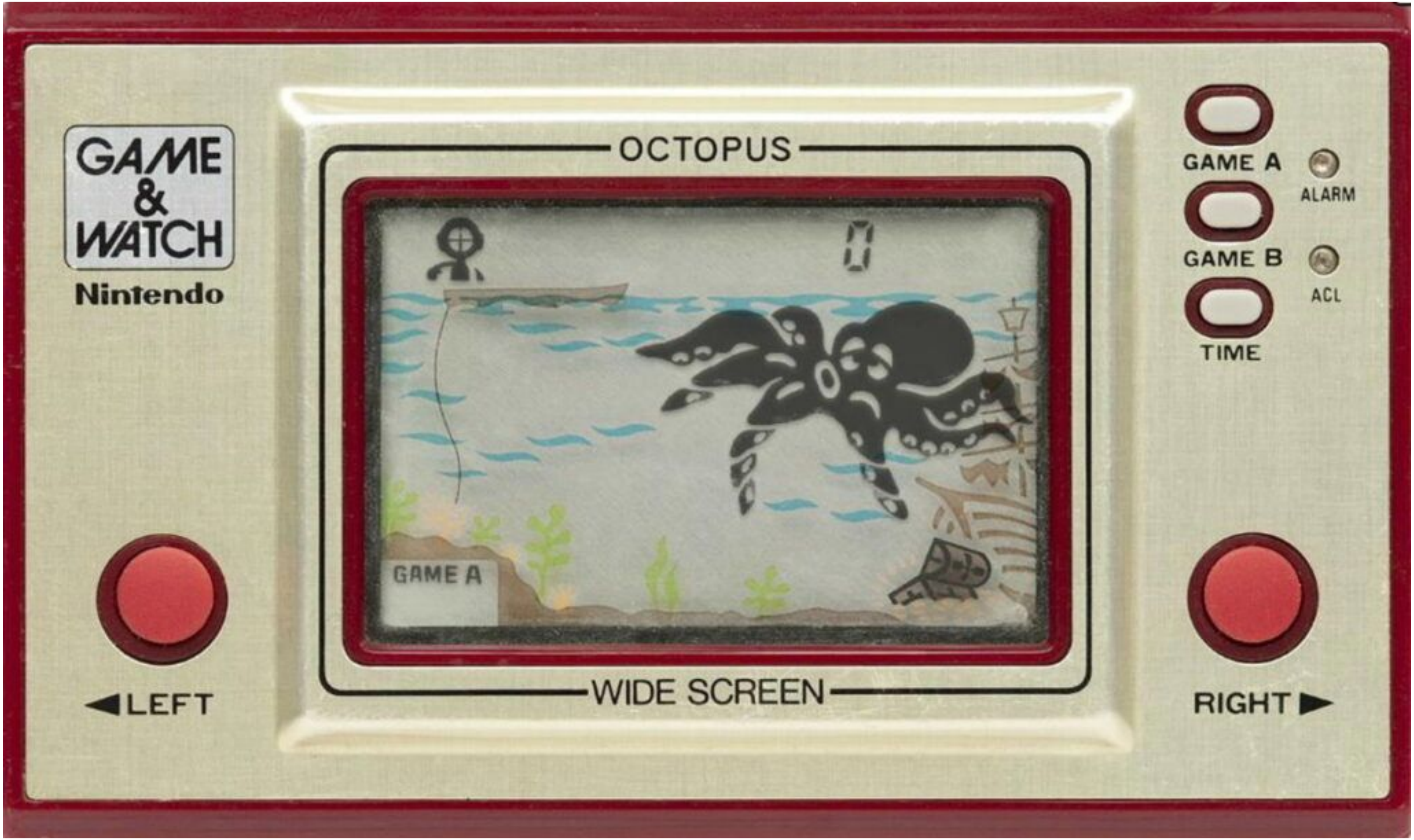
What is the best system to facilitate bone marrow and blood stem cell donations / transplants?

Common Systems

- Search Systems
- Recommendation Systems
- Referral Systems
- Affiliate Systems
- Notification Systems
- Gamification Systems
- Reward / Loyalty Systems
- Content Management Systems (CMS)
- Learning Management Systems (LMS)



What is the best technology to power the system?


GAME & WATCH
Nintendo
OCTOPUS
0
GAME A
WIDE SCREEN
GAME A
ALARM
GAME B
ACL
TIME
◀LEFT
RIGHT▶

Game and Watch

- **Inspiration:** Gunpei Yokoi observed a bored businessman playing with a calculator on a train,
- **Massive commercial success:** The series sold over 40 million units worldwide, helping establish Nintendo as a major player in the video game industry and paving the way for later handheld systems like the Game Boy.
- **Repurposing existing technology:** The device used low-cost LCD screens and chips originally developed for calculators.
- **Designer philosophy:** The system was designed by Gunpei Yokoi, who championed the idea of “lateral thinking with withered technology”—creating innovative products by creatively applying reliable, well-understood technology rather than chasing cutting-edge technologies.

Agents

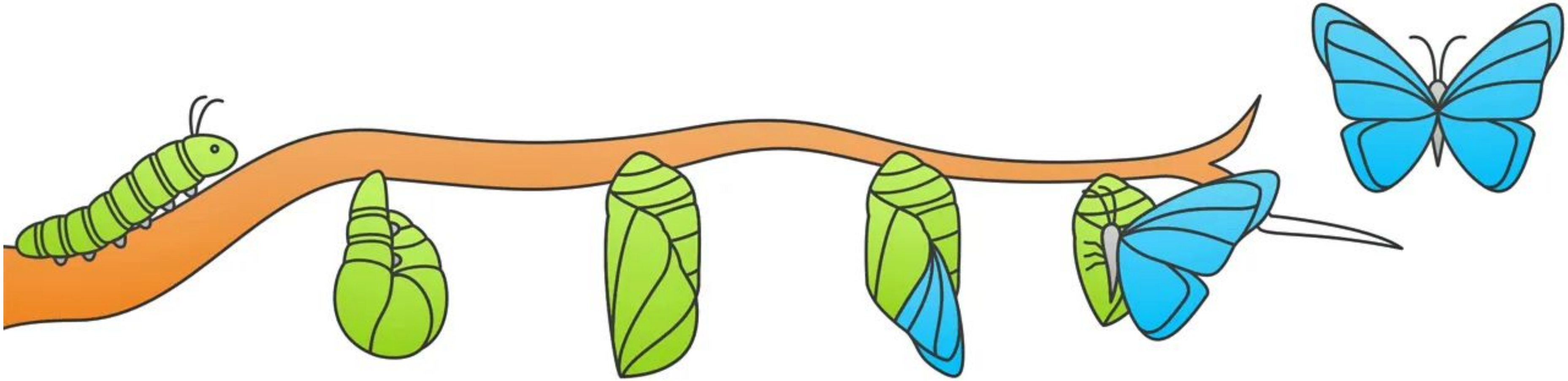
THE BEST INTERFACE IS NO INTERFACE

BY

GOLDEN KRISHNA

UX TIGERS

Evolution of AI Agents



Basic Autonomy
Limited task execution

Enhanced Autonomy
Improved task handling with user oversight

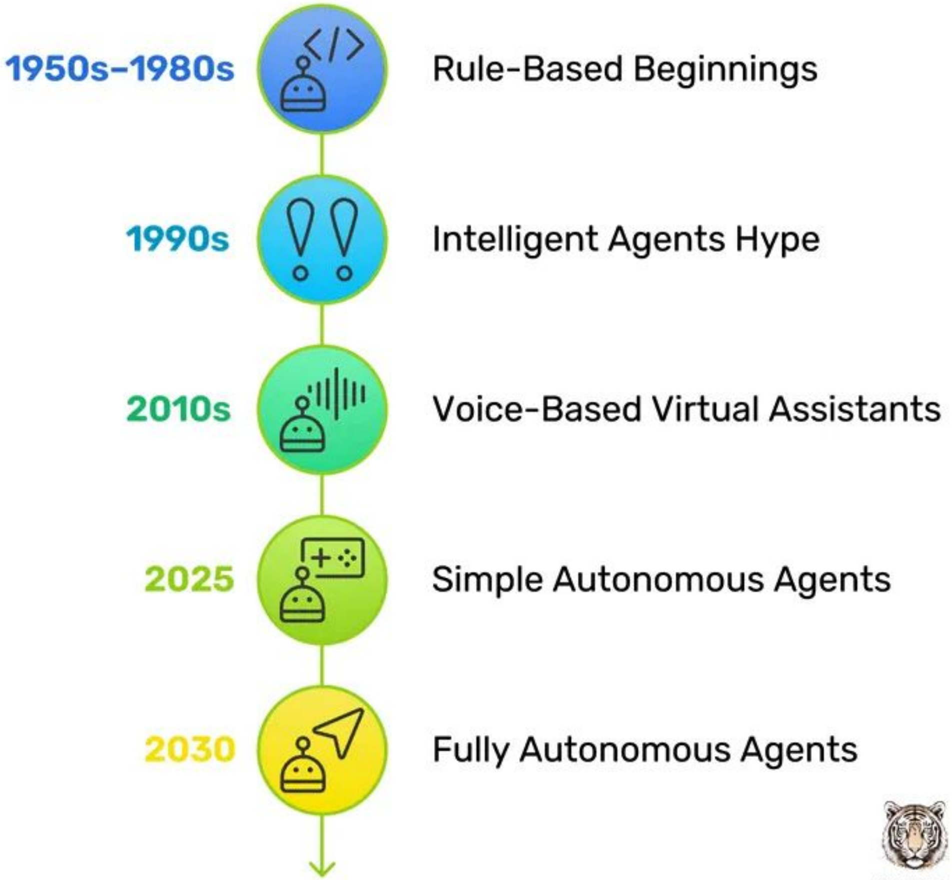
Multimodal Capabilities
Integration of text, images, and audio

Deep Integration
Seamless use within ecosystems and apps

Complex Workflows
Automation of multi-step processes

Advanced Assistants
High-level autonomy and productivity

Generations of AI Agents



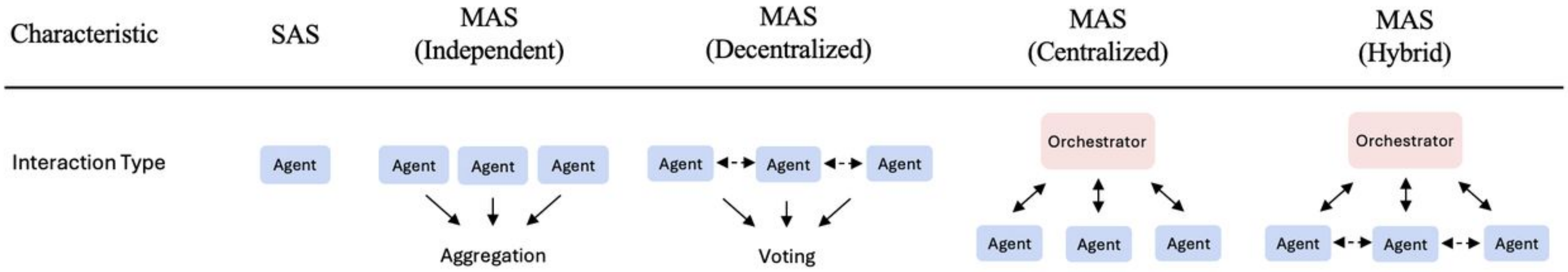
UX TIGERS

Hello AI Agents Article

- **AI agent** is a computer program that can sense information, make decisions, and act on its own to accomplish a task. It's an entity powered by artificial intelligence that operates with some independence, whether it's a virtual assistant on your phone (like Siri or Alexa), a character in a video game that acts intelligently, or a system to surf the web and take action on your behalf.
- **Use Cases:** Realistic use cases currently center on the digital realm: organizing information, automating online processes, analyzing data, and generating content. In the digital domain, AI agents do provide significant speed-ups (e.g., drafting a lengthy report would take humans hours, but an AI agent can assemble a decent draft in minutes). But they are not reliably better than humans at complex decision-making and are currently best used as tireless assistants to handle grunt work under supervision.

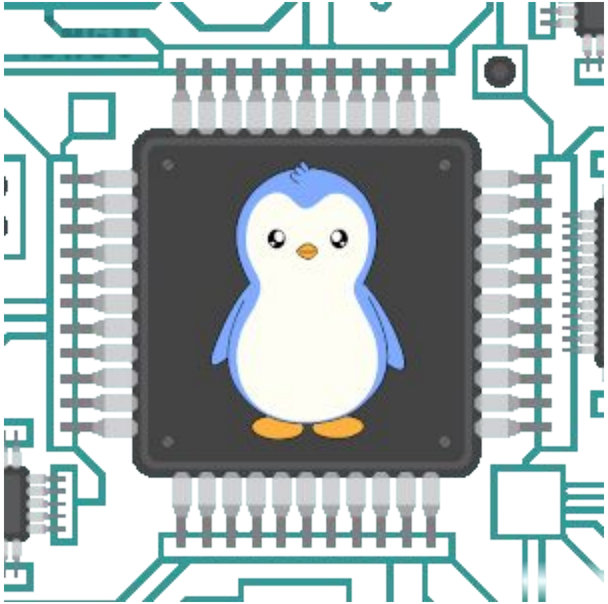
AI Agent Architecture

- **Single-Agent (SAS)**: A solitary agent executing all reasoning and acting steps sequentially with a unified memory stream.
- **Independent**: Multiple agents working in parallel on sub-tasks without communicating, aggregating results only at the end.
- **Centralized**: A "hub-and-spoke" model where a central orchestrator delegates tasks to workers and synthesizes their outputs.
- **Decentralized**: A peer-to-peer mesh where agents communicate directly with one another to share information and reach consensus.
- **Hybrid**: A combination of hierarchical oversight and peer-to-peer coordination to balance central control with flexible execution.

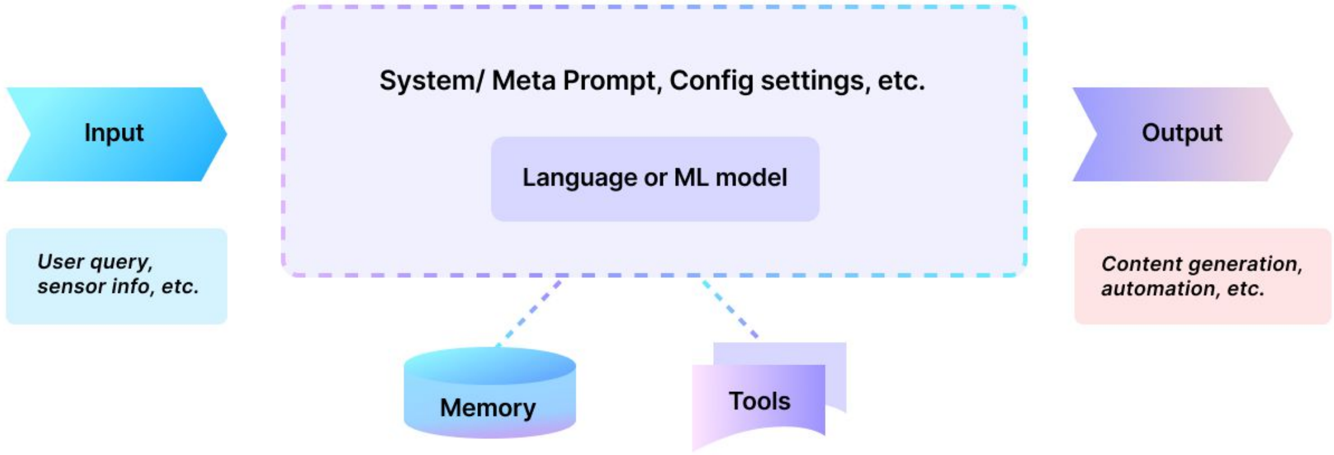


Key Questions

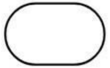
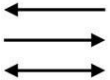
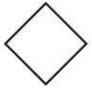
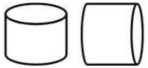
- Who will use it?
- Visible, invisible, and/or hybrid?
- What are its responsibilities?
- What kinds of inputs will it process?
- What data sources and knowledge bases will it need access to?
- What tools will it need access to?
- How will you measure success?



Software Architecture Diagramming



Software Architecture Diagramming

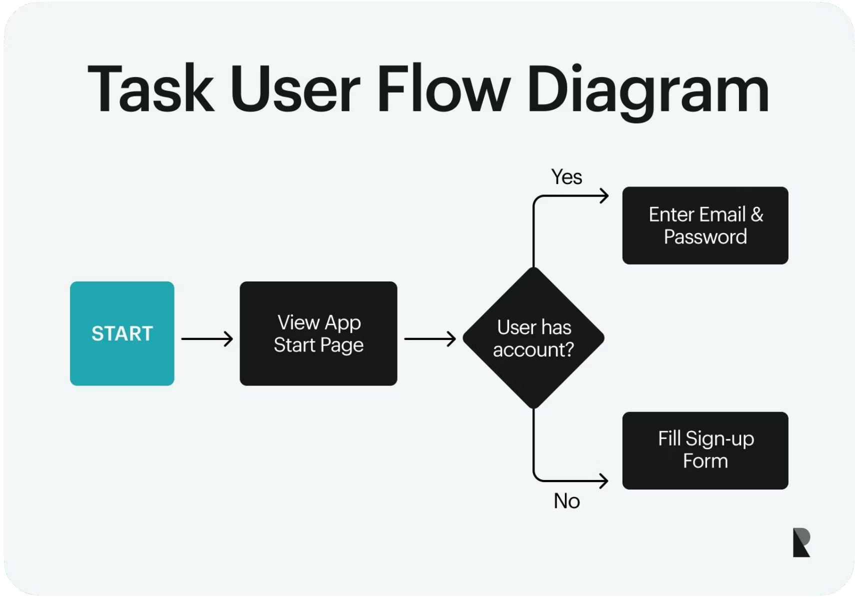
Name	Symbol	Function
Start / End		Used to represent the starting point or terminal point of a flowchart
Flow lines		Connects components in a flowchart and indicates flow direction
Input / Output		Represents information or data that is transmitted or received
Decision		Represents checkpoints to evaluate conditions for making decisions
Process		Represents processes (e.g., mathematical operations)
Database		Represents databases
Person		Represents actors or users or a software system

Task Flow Diagrams

Task Flow Diagrams

Like task flow diagrams, they describe what should happen but not how.Use case narratives follow a prescribed outline, specifying:

- Actors
- Pre-conditions
- Primary (“happy day”) flows
- Alternative flows
- Exception flows



Flow Diagramming Symbols for UX

Use Case Narratives

Like task flow diagrams, they describe what should happen but not how.Use case narratives follow a prescribed outline, specifying:

- Actors
- Pre-conditions
- Primary (“happy day”) flows
- Alternative flows
- Exception flows

Below is an example use case outline for cart checkout.

Use Case: Cart Checkout
Overview: Cart checkout is method customers can use to place an order
Actor: Customer
Pre-Conditions: The user has at least one item in the cart
Primary (Happy Day) Flow: The user is logged into their account
Alternative Flow A: The user is not logged into an account (guest checkout)
Exceptions: At any point up to placing the order, the user may cancel the checkout

Use Case Narrative + Task Flow Diagram

Cart Checkout

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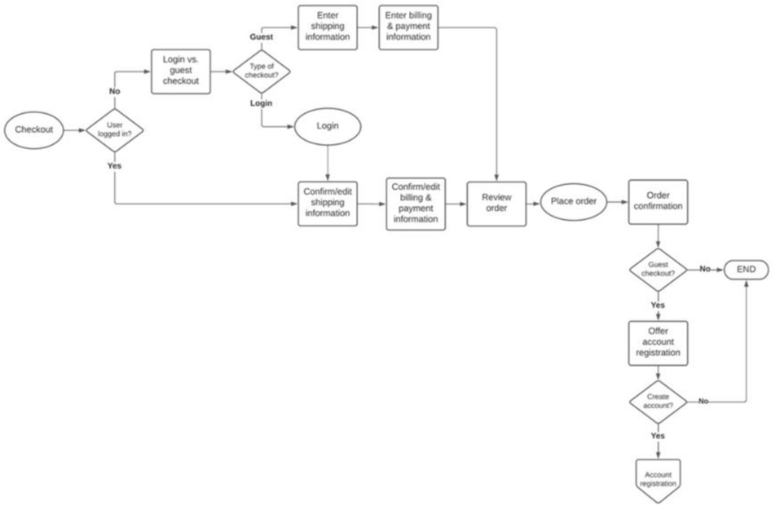


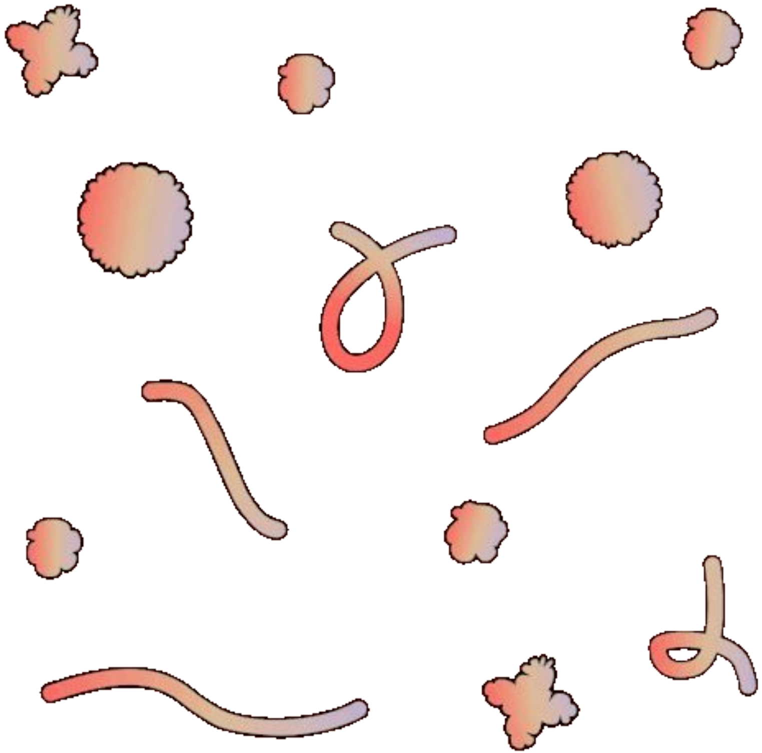
Figure 8.10
Task flow diagram with a use case outline. You can also view this diagram at objectmodelingfordesigners.com/chapter-8

Experiments

Product Experiments

- **Product experimentation** is the process of testing different hypotheses for product improvement through quantitative and qualitative data. These improvements can range from small design and color changes to new features.

- **Informal product experiments** are small-scale, rapid, and low-cost tests used to validate hypotheses, test ideas, or gauge interest before fully committing resources.



Thank you!